Griffiths 2021 Breast Cancer — analysis summary

RUN 4 OF 6

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

Harmony (θ=10) on sample integrates 67 samples across both technologies (10x+iCell8). Mixing judged on the per-neighbourhood iLISI (median 10.5/67), not the coarse same-sample scalar; continuity on the independent Harmony-PCA kNN graph (1 component). The manifold is one continuous, sample-mixed mass with a proliferative sub-region (not a confounded lineage); only ~1.4% detached specks removed → GO on the full manifold.

Gate 2 — module geography → GO (non-branching)

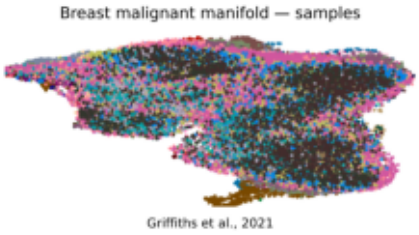
Organized cross-repression: smoothed pSom↔Crest r=−0.72 and Gast↔pSom r=−0.38 (raw pSom↔Crest −0.53). pSom and Crest occupy opposite territories on one continuous manifold — a single (Gast/pSom)→Crest inverse/terminal gradient, present but NOT resolved into two diverging arms → GO. No arm inversion to flag (Crest does not preferentially enrich a Gast arm).

Gate 3 — hub cross-repression → NA

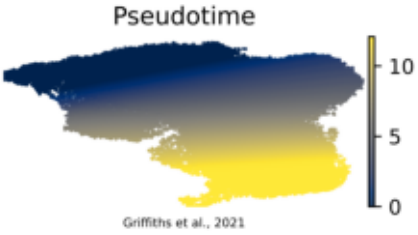
Non-branching topology (principal curve: 2 tips, 0 forks; fold-free self-approach 2.88). PPT 'forks' grow with node count — space-filling, not a real bifurcation. No hub bifurcation to test → NA.

STEP FIGURES

Manifold — samples



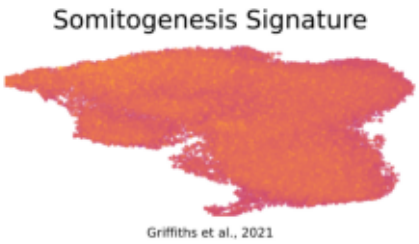
Pseudotime



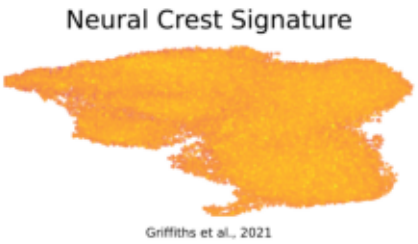
Gastrulation



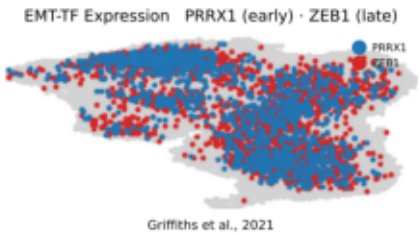
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Breast Cancer (ER+ neoadjuvant) (Griffiths et al., 2021), 10x + iCell8 — 108,658 malignant cells analysed (67 samples ≥100).

Verdict.
The embryonic pattern reproduces as a single gastrulation/somitogenesis → neural-crest axis (non-branching), with organized Gast/pSom→Crest cross-repression (pSom↔Crest r=−0.72) and a terminal Crest gradient; no Gast/pSom bifurcation, so Gate 3 is NA. Geometry Linear — consistent with the other CCCA breast datasets, on the full continuous manifold (no lobe removal). Gate 1 GO / Gate 2 GO / Gate 3 NA. Raw module means Gast −0.019 / pSom −0.053 / Crest +0.023. EMT-TFs PRRX1 (early) → ZEB1 (late).

Parameter choices

Threshold	≥100 malignant cells/sample (no dominant count cliff; median 646) → 67/95 samples, 108,658/111,372 cells	**Embedding**	Harmony θ=10 on sample (removes 10x/iCell8 batch); n_pcs=30, n_neighbors=30, spread=1.3; 4,072 overdispersed; no sc.pp.scale
Trajectory	scFates curve (non-branching), Nodes=20 λ=0.3 μ=100; fold-free (SA 2.88)	**Root**	Gast/pSom-high, Crest-low tip (Crest late/terminal); axis not proliferation-driven (cycling vs pt ρ=+0.03)
Geometry	Linear; raw means Gast −0.019 / pSom −0.053 / Crest +0.023	**EMT-TFs**	PRRX1 (early) + ZEB1 (late); all sparse (ZEB1 9%, rest <6%), pseudotime-flat